ECE 7410: Image Processing

Lab 1: Geometric Transformation and Histogram Equalization

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Geometric Transformation

1. Download the test image (im1.png) from Brightspace under Lab 01. Read the image, convert the image to a grayscale image, and display the image.

```
imc = imread("./im1.png");

img = rgb2gray(imc);

imshow(img);
title('Image in grayscale');
```

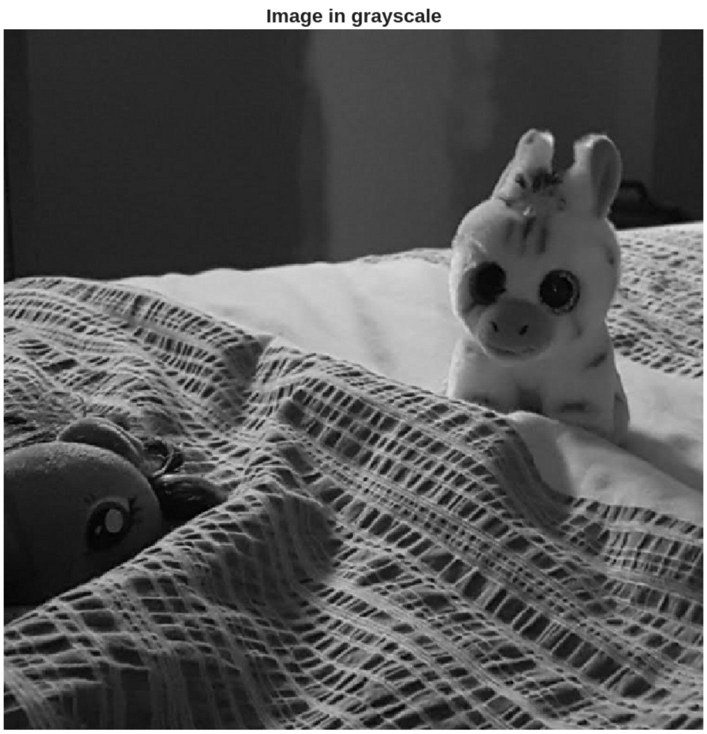

Figure 1: img1.png in greyscale

2. a. Define the transormation matrix with a rotation angle of 45 degrees

```
theta = 0.25*pi;
R = [cos(theta) sin(theta) 0;...
-sin(theta) cos(theta) 0; ...
0 0 1];
```

b. Calculate the size of the transformed image and generate an empty image with the calculated size

```
[y_max, x_max] = size(img); % Obtaining the size of the image
corners = [0,       0,      1; ...
           x_max,   0,      1; ...
           0,       y_max,  1; ...
           x_max,   y_max,  1]; % Corner points of the image

new_corners = corners*R % New location of corners
new_width = round(max(new_corners(:,1)) - ...
min(new_corners(:,1)));
new_height = round(max(new_corners(:,2)) - ...
min(new_corners(:,2)));
new_img = zeros(new_height,new_width);
```